

Question Bank

Chapter 1: Introduction to Cloud Computing and AWS

1. Define cloud computing and explain its core characteristics.
 2. What is meant by on-demand delivery of IT resources?
 3. Explain the pay-as-you-go pricing model in cloud computing.
 4. Differentiate between Public, Private, and Hybrid Clouds with examples.
 5. List and explain the three Cloud Service Models: IaaS, PaaS, and SaaS.
 6. Give AWS service examples for each cloud service model.
 7. Briefly describe the history of AWS.
 8. What are AWS Regions and Availability Zones?
 9. List four core AWS services and describe their purposes.
 10. What is IoT? How is it different from traditional computing?
 11. Give three real-world use cases of IoT applications.
 12. Mention two societal benefits of IoT technology.
 13. Discuss the privacy concerns associated with IoT.
 14. What security challenges does IoT bring?
 15. Name two AWS services used in IoT integration and their roles.
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Chapter 2: Networking Services

1. Define the terms IP, DNS, Subnet, and NAT.
2. What is a Virtual Private Cloud (VPC)? Why is it important?
3. List the core components of a VPC.
4. Explain the difference between public and private subnets.
5. How do route tables work within a VPC?
6. What is the role of Security Groups in AWS?
7. Differentiate between Security Groups and NACLs.

8. How are inbound and outbound rules configured in Security Groups?
 9. What is AWS Route 53? Mention its functions.
 10. Explain domain registration using Route 53.
 11. Describe routing policies in Route 53.
 12. What is AWS CloudFront? Why is it used?
 13. Define edge locations in the context of CloudFront.
 14. Explain how caching improves content delivery.
 15. Briefly outline the steps to build a VPC and launch an EC2 instance inside it.
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Chapter 3: Compute Services

1. What is Infrastructure as a Service (IaaS)? Give an AWS example.
 2. Describe the process to launch, configure, and terminate an EC2 instance.
 3. List and explain types of Elastic Load Balancer (ELB) in AWS.
 4. Give one use case for each type of ELB.
 5. What is Auto Scaling and why is it important?
 6. Explain Auto Scaling policies and scaling plans.
 7. Define serverless computing with an example.
 8. What is AWS Lambda and how does it work?
 9. Describe the event-driven model in AWS Lambda.
 10. List three use cases of AWS Lambda.
 11. What is AWS Elastic Beanstalk? Why is it considered a PaaS?
 12. Explain the steps to deploy an application using Elastic Beanstalk.
 13. How does Auto Scaling integrate with Elastic Load Balancing?
 14. What are the benefits of serverless architectures?
 15. Differentiate between IaaS and PaaS with AWS services as examples.
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Chapter 4: Storage Services

1. Differentiate between Object, Block, and File storage.
 2. What is Amazon S3? Mention its core components.
 3. Explain the various storage classes available in S3.
 4. What is versioning in S3? Why is it useful?
 5. How do Lifecycle policies in S3 work?
 6. What is Amazon EBS and when should it be used?
 7. Describe the process of creating and attaching an EBS volume.
 8. What is a snapshot in EBS? Why is it useful?
 9. Define Amazon EFS and its use cases.
 10. How does EFS differ from EBS?
 11. Explain how to host a static website using S3.
 12. List the required permissions to host a static website on S3.
 13. Compare S3, EBS, and EFS based on use case.
 14. Discuss the cost implications of using S3, EBS, and EFS.
 15. What storage option is ideal for a media streaming service and why?
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Chapter 5: AWS Architecture and Database Services

1. What is the AWS Well-Architected Framework?
2. List and explain its five design principles.
3. Define Relational Database Service (RDS).
4. Which database engines are supported by RDS?
5. Explain the steps to launch and connect to an RDS instance.
6. What is Amazon Aurora and how is it different from RDS?
7. Mention three benefits of using Amazon Aurora.
8. What is DynamoDB? List its key features.
9. How does DynamoDB differ from RDS?

10. State three use cases for DynamoDB.
 11. What is Amazon Redshift used for?
 12. Describe the process of launching an Amazon RDS DB instance.
 13. Compare DynamoDB and Redshift based on use case.
 14. Give a scenario where Aurora would be a better choice over standard RDS.
 15. Draw a simple AWS architecture diagram for a web application.
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Chapter 6: AWS Security, Monitoring, Scaling, and Billing

1. Define IAM and list its core components.
 2. Explain the principle of least privilege in IAM.
 3. What are IAM roles and how do they differ from users?
 4. How are IAM policies created and attached?
 5. List two best practices for IAM security.
 6. What is AWS CloudWatch? What are its main features?
 7. How are CloudWatch alarms configured?
 8. Define metrics in the context of CloudWatch.
 9. What is the role of Elastic Load Balancing in AWS?
 10. How does Auto Scaling work with CloudWatch alarms?
 11. What is the purpose of AWS Cost Explorer?
 12. How are AWS Budgets different from billing alarms?
 13. Explain the role of the AWS Pricing Calculator.
 14. What is AWS Config and how does it support compliance?
 15. Describe AWS CloudTrail and its function.
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